Dixin Tang dixin [at] utexas [dot] edu

Research Areas

Vector Databases, Data Agents, Data Systems for AI Agents

Employment

2024-present	Assistant Professor - The University of Texas at Austin
2021-2023	Postdoctoral Scholar - University of California, Berkeley Advisor: Aditya G. Parameswaran, Associate Professor

Education

2015-2020	Ph.D. in Computer Science - University of Chicago Advisor: Aaron J. Elmore, Associate Professor
2011-2014	M.S. in Computer Science - Institute of Computing Technology, Chinese Academy of Sciences Advisor: Wei Li, Associate Professor
2007-2011	B.S. in Software Engineering - Huazhong University of Science & Technology

Publications

P23. Rewinding the Fold: Making Stateful Computations Retractable
Ziteng Wang, Linux Zheng, **Dixin Tang**, Isil Dillig
POPL 2027

P22. Optimal Predicate Pushdown Synthesis
Robert Zhang, Eric Hayden Campbell, **Dixin Tang**, Isil Dillig
PLDI 2026

P21. Homomorphism Calculus for User-Defined Aggregations
Ziteng Wang, Ruijie Fang, Linus Zheng, **Dixin Tang**, Isil Dillig
OOPSLA 2025

P20. Transactional Panorama: A Conceptual Framework for User Perception in Analytical Visual Interfaces (extended version)
Dixin Tang, Alan Fekete, Indranil Gupta, Aditya G. Parameswaran
VLDBJ 2025 (The **"Best of VLDB 2023"** special issue of VLDB Journal)

P19. Tigon: A Distributed Database for a CXL Pod
Yibo Huang, Haowei Chen, Newton Ni, Yan Sun, Vijay Chidambaram, **Dixin Tang**, Emmett Witchel
OSDI 2025

P18. Impeller: Stream Processing on Shared Logs
Zhiting Zhu, Zhipeng Jia, Newton Ni, **Dixin Tang**, Emmett Witchel
EuroSys 2025

P17. Pasha: An Efficient, Scalable Database Architecture for CXL Pods
Yibo Huang, Newton Ni, Vijay Chidambaram, Emmett Witchel, **Dixin Tang**
CIDR 2025

P16. Dealing with Acronyms, Abbreviations, and Typos in Real-World Entity Matching
Joshua Wu, **Dixin Tang**, Nithin Chalapathi, Tristan Chambers, Julie Ciccolini, Cheryl Phillips, Lisa Pickoff-White, Aditya G. Parameswaran
VLDB 2024, Industrial Track

P15. Visualizing Spreadsheet Formula Graphs Compactly
Fanchao Chen, **Dixin Tang**, Haotian Li, Aditya G. Parameswaran
VLDB 2023, Demo

P14. Transactional Panorama: A Conceptual Framework for User Perception in Analytical Visual Interfaces
Dixin Tang, Alan Fekete, Indranil Gupta, Aditya G. Parameswaran
VLDB 2023 (**"Best of VLDB 2023"**)

P13. Efficient and Compact Spreadsheet Formula Graphs
Dixin Tang, Fanchao Chen, Christopher De Leon, Tana Wattanawaroon, Jeaseok Yun, Srinivasan Seshadri, Aditya G. Parameswaran
ICDE 2023

P12. Flexible Rule-Based Decomposition and Metadata Independence in Modin: A Parallel Dataframe System
Devin Petersohn*, **Dixin Tang***, Rehan Durrani, Areg Melik-Adamyan, Joseph E. Gonzalez, Anthony D. Joseph, Aditya G. Parameswaran
VLDB 2022 (*Equal contribution)

P11. Lux: Always-on Visualization Recommendations for Exploratory Dataframe Workflows
Doris Jung-Lin Lee, **Dixin Tang**, Kunal Agarwal, Thyne Boonmark, Caitlyn Chen, Jake Kang, Ujjaini Mukhopadhyay, Jerry Song, Micah Yong, Marti A. Hearst, Aditya G. Parameswaran
VLDB 2022

P10. Enhancing the Interactivity of Dataframe Queries by Leveraging Think Time
Doris Xin, Devin Petersohn, **Dixin Tang**, Yifan Wu, Joseph E. Gonzalez, Joseph M. Hellerstein, Anthony D. Joseph, Aditya G. Parameswaran
IEEE Data Eng. Bull. 2021

P9. Resource-Efficient Shared Query Execution via Exploiting Time Slackness
Dixin Tang, Zechao Shang, William Ma, Aaron J. Elmore, Sanjay Krishnan
SIGMOD 2021

P8. CIAO: An Optimization Framework for Client-Assisted Data Loading
Cong Ding, **Dixin Tang**, Xi Liang, Aaron J. Elmore, Sanjay Krishnan
ICDE 2021, Short Paper

P7. CrocodileDB in Action: Resource-Efficient Query Execution by Exploiting Time Slackness
Dixin Tang, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin
VLDB 2020, Demo

P6. Thrifty Query Execution via Incrementability
Dixin Tang, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin
SIGMOD 2020

P5. CrocodileDB: Efficient Database Execution through Intelligent Deferment
Zechao Shang, Xi Liang, **Dixin Tang**, Cong Ding, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin
CIDR 2020

P4. Intermittent Query Processing
Dixin Tang, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin
VLDB 2019

P3. Socrates: The New SQL Server in the Cloud
Panagiotis Antonopoulos, Alex Budovski, Cristian Diaconu, Alejandro Hernandez Saenz, Jack Hu, Hanuma Kodavalla, Donald Kossmann, Umar Farooq Minhas, Naveen Prakash, Hugh Qu, Chaitanya Sreenivas Ravella, Krystyna Reisteter, Sheetal Shroti, **Dixin Tang**, Vikram Wakade
SIGMOD 2019

P2. Toward Coordination-Free and Reconfigurable Mixed Concurrency Control
Dixin Tang, Aaron J. Elmore
USENIX'ATC 2018

P1. Adaptive Concurrency Control: Despite the Looking Glass, One Concurrency Control Does Not Fit All
Dixin Tang, Hao Jiang, Aaron J. Elmore
CIDR 2017

Team

PhD Students

- Abhinav Gyawali (co-advised with Prof. Chidambaram)
- Hexu Li
- Shurang Wu (co-advised with Prof. Chidambaram)
- Tianming Wu
- Yibo Huang (co-advised with Prof. Witchel)
- Ziteng Wang (co-advised with Prof. Dillig)

Undergraduate and Master's Students

- Yingfei Tang

Honors and Awards

- Distinguished Reviewer Award, SIGMOD 2025
- Best of VLDB 2023

Professional Services

Program Committee

- SIGMOD - 2022, 2023, 2025, 2026, 2027, 2028
- VLDB - 2025, 2026, 2027
- ICDE - 2025, 2026
- EDBT - 2024
- SIGMOD Demo Track - 2022, 2024, 2025
- HILDA - 2023, 2024

Conference Reviewer

- IEEE VIS - 2021

Journal Reviewer

- VLDB Journal
- Distributed and Parallel Databases Journal

Teaching

The University of Texas at Austin

- Instructor - CS 395T: Advanced Database Systems, Fall 2026
- Instructor - CS 378: Database System Design, Spring 2026
- Instructor - CS 395T: Database Systems and LLMs, Fall 2025
- Instructor - CS 347: Data Management, Fall 2024
- Instructor - CS 395T: Advanced Query Optimizations, Spring 2024

University of Chicago

- TA - CMSC 23500 - Introduction to Database Systems, Winter 2017-2020
- TA - MPCS 52040 - Distributed Systems, Spring 2016
- TA - MPCS 51040 - C Programming, Fall 2015

Industry Experience

Internship at Microsoft Research June 2018-Sep. 2018

Project: Benchmarking Socrates *Mentor:* Umar Farooq Minhas

Socrates is a new cloud-native database that decouples computation from storage. My internship involved testing the new database architecture of Socrates in an industrial setting, understanding its performance bottlenecks, and proposing optimization opportunities.